# EX-GIRLFRIEND

## USER MANUAL




Four disting EX-style instruments: **EX Multisample** · **EX Poly Wavetable** · **EX Poly FM** · **EX Poly Exciter**

VST3 · AU · LV2 · standalone - macOS, Windows and Linux

Version 1.0.0 - github.com/femifleming/ex-girlfriend

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# INTRODUCTION

**ex-girlfriend** is a suite of four software instruments modelled on algorithms of the Expert Sleepers **disting EX** Eurorack module. They follow the module's manual (firmware 1.26) parameter for parameter, use the same file formats (sample folders, wavetables, exciters, DX7 banks, Scala tunings) and wear the module's look: the whole window is one white-on-black pixel display in the module's own 5×7 font.

| Plug-in | Algorithm(s) | In short |
|---|---|---|
| EX Multisample | 3 - SD Multisample<br>5 - WAV Recorder | 8-voice, 3-timbre sample player with velocity layers, round robins, chords, arpeggiator, EQ and delay; plus a recorder with an **auto-sampler** that samples any MIDI/CV instrument into a ready-to-play folder. |
| EX Poly Wavetable | 10 - Poly Wavetable | 8-voice wavetable synth: 2 envelopes, filter, LFO, unison, Juno-style chorus and delay. Imports WaveEdit banks. |
| EX Poly FM | 23 - Poly FM | 8-voice, 4-timbre 6-operator FM (the Plaits engine). Loads DX7 .syx banks; DX7 editors can edit it over SysEx. |
| EX Poly Exciter | 13 - Poly Exciter | 8-voice exciter/resonator synth: plucked, struck and bowed-like sounds from short exciter samples or live input. |

Each instrument is available as a **VST3** (all platforms), **Audio Unit** (macOS), **LV2** (Linux) and a **standalone app**. Settings are saved with your DAW project, and every plug-in also remembers its last state between sessions.

ex-girlfriend is not affiliated with or endorsed by Expert Sleepers; "disting" is their trademark and is used only to describe compatibility. If you own the module, its MicroSD card works directly with these plug-ins.

# INSTALLING

All downloads are on the release page: **github.com/femifleming/ex-girlfriend/releases**.

## MACOS - HOMEBREW

```
brew tap femifleming/ex-girlfriend https://github.com/femifleming/ex-girlfriend
brew trust femifleming/ex-girlfriend
brew install --cask ex-girlfriend            # apps + VST3 + AU
brew install --cask ex-girlfriend-sd-card    # optional sample library
```

Homebrew 7 asks you to *trust* third-party taps once. To build from source instead (macOS or Linux): `brew install ex-girlfriend`, then follow the link instructions it prints.

## MACOS - DOWNLOAD

- **ex-girlfriend-1.0.0-macOS.zip**: unzip and double-click **install.command** - apps go to ~/Applications, plug-ins to ~/Library/Audio/Plug-Ins/VST3 and /Components.
- **ex-girlfriend-1.0.0-macOS.pkg**: installs for all users (right-click -> Open the first time).
- The builds are universal (Apple silicon + Intel) but not notarised. If macOS or your DAW refuses to open them, run:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/VST3/EX*.vst3 \
    ~/Library/Audio/Plug-Ins/Components/EX*.component ~/Applications/EX*.app
```

## WINDOWS

Unzip **ex-girlfriend-1.0.0-Windows-x64.zip**, right-click **install.ps1** -> *Run with PowerShell* (as administrator). VST3 plug-ins go to C:\Program Files\Common Files\VST3; the apps appear in the Start menu under **ex-girlfriend**.

## LINUX

```
tar xzf ex-girlfriend-1.0.0-Linux-x86_64.tar.gz
cd ex-girlfriend-1.0.0-Linux-x86_64 && ./install.sh
```

VST3 -> ~/.vst3, LV2 -> ~/.lv2, apps -> ~/.local/bin with desktop menu entries.

## THE SAMPLE LIBRARY

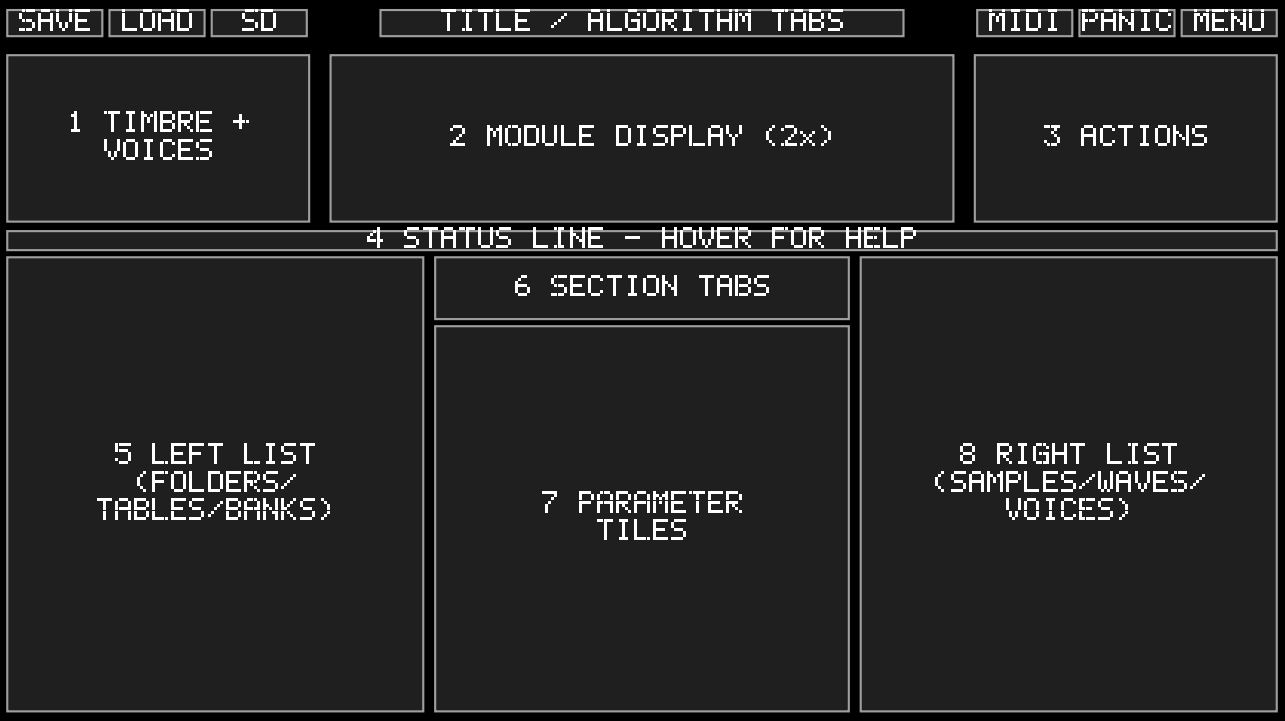
Download **EX-SD-Card.zip** and unzip it into your **Music** folder so you have **Music/EX SD Card** (the default location), or pick any folder with the **SD** button. It holds 18 multisampled instruments, 38 wavetables, 5 exciter tables, DX7 banks and Scala tunings - all free (CC0 / CC-BY / MIT, see CREDITS.txt inside). A real disting EX MicroSD card works too.

## BUILDING FROM SOURCE

```
git clone --recursive https://github.com/femifleming/ex-girlfriend
cd ex-girlfriend
cmake -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build --config Release
```

# THE SCREEN

Every plug-in uses the same mirror-symmetrical layout. The window can be resized; it snaps to whole-pixel sizes (1×, 2×, 3× …) so the pixels stay sharp.




| | Area | What it does |
|---|---|---|
| | Top bar | **SAVE / LOAD** presets (stored on the SD card), **SD** chooses the card folder, **MIDI** channels, **PANIC** (all notes off), **MENU** (every option of the module's menus). In EX Multisample the centre holds the algorithm tabs. |
| 1 | Timbre / voices | Timbre buttons (multi-timbral plug-ins) choose which timbre the lists edit. The 8 cells show each voice's note and level. |
| 2 | Module display | The disting EX's own 128×32 screen: current parameter, live voices, levels, auto-sampler progress. Mouse wheel = parameter, shift+wheel = value; Up/Down and Left/Right act like the P and V encoders, Enter = push P (menu), Esc = push V. |
| 3 | Actions | Buttons for the algorithm: RECORD, PLAY, AUTO, TEST, IMPORT, WAVEEDIT, NOTES OFF, RESCAN … |
| 4 | Status line | Hover anything for a description. MIDI IN / OUT activity lights sit at either end. |
| 5/8 | Lists | Click to load: sample folders and their samples, wavetables and their waves, exciter tables and exciters, FM banks and voices, recordings. |
| 6 | Section tabs | Pages of related parameters (e.g. ENV, FILTER, CHORD, FX). |
| 7 | Parameter tiles | Drag up/down or use the wheel to change (hold shift for fine steps); double-click resets to the default. Tiles with a small (v) open a list of choices when clicked. |

## PLAYING NOTES

Play from MIDI, or click the window and use the computer keyboard: **A W S E D F T G Y H U J K O L P ;** play a chromatic octave and a half, **Z**/**X** shift the octave.

## MIDI

- The receive channel defaults to 1 (Omni is available under MIDI). Multi-timbral plug-ins listen on consecutive channels (timbre 1 on the receive channel, timbre 2 on the next …) unless you assign each timbre its own channel.
- Recognised: note on/off, velocity, pitch bend, sustain (CC64), All Notes Off (CC123); channel and poly aftertouch in Poly Exciter; DX7 SysEx in Poly FM. *Delay uses MIDI clk* (menu) locks delay times to the host tempo.

## OUTPUTS

**Out 1/2** is the main stereo output. **Out 3/4** is a second stereo bus (enable it in your DAW) carrying per-timbre or per-voice outputs depending on the *Output mode*, the paraphonic gate and mono mix (as on the module), and the auto-sampler's CV/gate.

## SAVING AND REMEMBERING

- In a DAW, each instance is saved with the project.
- **SAVE** writes a preset to *SD card/presets/<plug-in>*; **LOAD** lists them (plus *Reset to defaults*).
- Every plug-in and app also remembers its **last state**: it is written whenever you change something and when the plug-in closes, and a new instance starts from it. The files live in ~/Library/Application Support/ex-girlfriend (macOS), %APPDATA%\ex-girlfriend (Windows) or ~/.config/ex-girlfriend (Linux).

# THE SD CARD FOLDER

The plug-ins read a folder laid out exactly like the disting EX's MicroSD card (default **Music/EX SD Card**).

| Folder | Contents |
|---|---|
| <any name>/ | A **multisample folder** of WAV files (see naming below). Every top-level folder that contains WAVs is listed. |
| wavetables/ | Each wavetable is a sub-folder of single-cycle WAVs (used alphabetically), or one WAV of concatenated cycles (600 samples each; 256 for WaveEdit banks - detected automatically). Optional *playlist-wavetable.txt* with -wavelength=. |
| exciters/ | Each exciter table is a folder of up to 128 short WAVs, or one WAV containing exactly 128 equal-length exciters. |
| FMSYX/ | DX7 32-voice bank SysEx files (.syx, 4104 bytes). Without any, three built-in banks are used. |
| scl/ kbm/ | Scala scales and keyboard maps for the Scala/MTS and Scala KBM parameters. |
| !RECORD/ | WAV Recorder recordings (REC00001.WAV …). |
| AUTO0001/ … | Auto-sampler output - ordinary multisample folders. |
| presets/ | Presets saved with SAVE. |

## NAMING MULTISAMPLES

| Token | Example | Meaning |
|---|---|---|
| _<note> | piano_A2.wav, piano_C#5.wav | Natural pitch. Sharps only, C4 = MIDI note 60. |
| _SW<n> | piano_A2_SW40.wav | Switch point: the lowest MIDI note that uses this sample. |
| _V<n> | snare_V1.wav, snare_V2.wav | Velocity layer; V1 is the softest. |
| _RR<n> | snare_RR1.wav, snare_RR2.wav | Round-robin variants of the same sample. |
| combined | cello_C3_V2_RR1.wav | Velocity first, then round robin. |

- Files without a note name are mapped chromatically from note 48 (C3) upwards - handy for one-shots.
- Without explicit switch points, gaps of up to 3 semitones are covered by pitching the upper sample down; for larger gaps the lower sample is also pitched up over part of the gap (exactly as on the module).
- Loops are read from the WAV's *smpl* or *cue* chunks. A sample loops if it has loop points, otherwise it plays once.
- An optional *playlist.txt* ("disting playlist v1") can set -natural=, -switch=, -loop= and -endLoopOnRelease= per file.

# EX MULTISAMPLE




## 3 - SD MULTISAMPLE

An 8-voice, 3-timbre sample instrument. Choose a folder in the left list; the right list shows its samples - click one to hear it. The display's top line shows the file, natural note and switch point of the highest playing voice.

- **Timbres**: the TIMBRE buttons pick which timbre the folder list sets. Timbres 2 and 3 can follow timbre 1 or use their own folders, and each has its own note range and arpeggiator.
- **SOUND**: gain, saturation, voice count, round-robin mode, output mode and stereo spread.
- **ENV**: attack/decay/sustain/release; release 127 = infinite (the sample plays to its end or loops forever).
- **TUNE**: octave, transpose, fine tune, bend range, Scala tuning and per-voice detune (an "old polysynth" spread).
- **CHORD**: chord generator (key, scale, shape, inversion), arpeggiator per timbre and broken chords (strum time/direction).
- **FX**: stereo or ping-pong delay and bass/treble EQ. **KEYS**: sustain behaviour, velocity curve and note ranges.
- MENU -> SD Multisample -> Options: MIDI note range, paraphonic gate mode, *Delay uses MIDI clk*, *Transpose limit*.

### CHORDS AND ARPEGGIOS

| Scales | Shapes | Arpeggio modes |
|---|---|---|
| Major, Natural Minor, Dominant, Fully Diminished, Dominant Dim, Augmented, Whole Tone, Chromatic, Harmonic Minor | None, Octave, Two Octaves, Root/Fifth, Root/Fifth + 8ve, Triad, Triad + 8ve, Sus4, Sus4 + 8ve, Sixth, Sixth + 8ve, Seventh, Seventh + 8ve, Ninth | Off, Up, Down, Alt, Alt2, Up/Down/Alt/Alt2 -8ve (skip the repeated octave), Step Up, Step Down, Random, Random2 (no repeats), Random3 (permutations), As Played |

Each new note (gate) advances the arpeggio by one step through the chord built on it; with chords off, the arpeggio runs over the notes you are holding. *Range* 2 or 3 adds copies one or two octaves up.

# PARAMETERS

| # | Parameter | Range | Default | Page | Values / notes |
|---|---|---|---|---|---|
| 1 | Attenuverter 1 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 2 | Attenuverter 2 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 3 | Attenuverter 3 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 4 | Attenuverter 4 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 5 | Attenuverter 5 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 6 | Attenuverter 6 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 7 | Folder | 0 .. 999 | 0 | list | |
| 8 | Attack time | 0 .. 127 | 0 | ENV | |
| 9 | Decay time | 0 .. 127 | 60 | ENV | |
| 10 | Sustain level | 0 .. 127 | 127 | ENV | |
| 11 | Release time | 0 .. 127 | 77 | ENV | |
| 12 | Octave | -8 .. 8 | 0 | TUNE | |
| 13 | Transpose | -60 .. 60 | 0 | TUNE | |
| 14 | Fine tune | -100 .. 100 cents | 0 cents | TUNE | |
| 15 | Gain | -40 .. 24 dB | 0 dB | SOUND | |
| 16 | Saturation | 0 .. 1 | On | SOUND | 0=Off, 1=On |
| 17 | Sustain | 0 .. 1 | Off | KEYS | 0=Off, 1=On |
| 18 | Max voices | 1 .. 8 | 8 | SOUND | |
| 19 | Bend range | 0 .. 48 | 2 | TUNE | |
| 20 | Pitch bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 21 | Voice 1 detune | -100 .. 100 cents | 0 cents | TUNE | |
| 22 | Voice 2 detune | -100 .. 100 cents | 0 cents | TUNE | |
| 23 | Voice 3 detune | -100 .. 100 cents | 0 cents | TUNE | |
| 24 | Voice 4 detune | -100 .. 100 cents | 0 cents | TUNE | |
| 25 | Voice 5 detune | -100 .. 100 cents | 0 cents | TUNE | |
| 26 | Voice 6 detune | -100 .. 100 cents | 0 cents | TUNE | |
| 27 | Voice 7 detune | -100 .. 100 cents | 0 cents | TUNE | |
| 28 | Voice 8 detune | -100 .. 100 cents | 0 cents | TUNE | |
| 29 | Chord enable | 0 .. 7 | Off | CHORD | 0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3 |
| 30 | Chord key | -12 .. 12 | 0 | CHORD | |
| 31 | Chord scale | 0 .. 8 | Major | CHORD | 0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic, 8=Harmonic Minor |
| 32 | Chord shape | 0 .. 13 | None | CHORD | 0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth |
| 33 | Chord inversion | 0 .. 3 | 0 | CHORD | |
| 34 | Arpeggio 1 mode | 0 .. 14 | Off | CHORD | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |
| 35 | Arpeggio 2 mode | 0 .. 14 | Off | CHORD | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |

| # | Parameter | Range | Default | Page | Values / notes |
|---|---|---|---|---|---|
| 36 | Arpeggio 3 mode | 0 .. 14 | Off | CHORD | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |
| 37 | Arpeggio 1 range | 1 .. 3 | 1 | CHORD | |
| 38 | Arpeggio 2 range | 1 .. 3 | 1 | CHORD | |
| 39 | Arpeggio 3 range | 1 .. 3 | 1 | CHORD | |
| 40 | Scala/MTS | -16 .. 99 | 0 | TUNE | |
| 41 | Scala KBM | 0 .. 99 | 0 | TUNE | |
| 42 | Folder 2 | -1 .. 999 | -1 | list | |
| 43 | Folder 3 | -1 .. 999 | -1 | list | |
| 44 | Min note 1 | 0 .. 127 | 0 | KEYS | |
| 45 | Max note 1 | 0 .. 127 | 127 | KEYS | |
| 46 | Min note 2 | 0 .. 127 | 0 | KEYS | |
| 47 | Max note 2 | 0 .. 127 | 127 | KEYS | |
| 48 | Min note 3 | 0 .. 127 | 0 | KEYS | |
| 49 | Max note 3 | 0 .. 127 | 127 | KEYS | |
| 50 | Output spread | -100 .. 100 % | 0 % | SOUND | |
| 51 | Delay mode | 0 .. 2 | Off | FX | 0=Off, 1=Stereo, 2=Ping-pong |
| 52 | Delay level | -40 .. 0 dB | -3 dB | FX | |
| 53 | Delay time | 1 .. 1365 ms | 500 ms | FX | |
| 54 | Delay feedback | 0 .. 100 % | 50 % | FX | |
| 55 | Tone bass | -240 .. 240 | 0 | FX | |
| 56 | Tone treble | -240 .. 240 | 0 | FX | |
| 57 | Break time | 0 .. 1000 ms | 0 ms | CHORD | |
| 58 | Break direction | 0 .. 2 | Up | CHORD | 0=Up, 1=Down, 2=Alternate |
| 59 | Voice 1 bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 60 | Voice 2 bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 61 | Voice 3 bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 62 | Voice 4 bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 63 | Voice 5 bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 64 | Voice 6 bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 65 | Voice 7 bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 66 | Voice 8 bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 67 | Output mode | 0 .. 4 | All summed 1 | SOUND | 0=All summed 1, 1=All summed 2, 2=Per gate 12-34, 3=Per gate 12-34 2, 4=Per gate 1-2-3 |
| 68 | Input mode | 0 .. 9 | 1 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 69 | Sustain mode | 0 .. 1 | Synth | KEYS | 0=Synth, 1=Piano |
| 70 | MIDI vel curve | 0 .. 3 | Linear | KEYS | 0=Linear, 1=Soft, 2=Hard, 3=Fixed |
| 71 | Arp reset input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 72 | Gate offset | 0 .. 496 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 73 | Round robin mode | 0 .. 3 | Sequential | SOUND | 0=Sequential, 1=Random, 2=Random2, 3=Random3 |




## 5 - WAV RECORDER AND AUTO-SAMPLER

Switch to the recorder with the **5 WAV RECORDER** tab. **RECORD** records the plug-in's audio input to *!RECORD/RECnnnnn.WAV* (48 or 96 kHz, 16 or 24-bit, mono or stereo); **PLAY** plays the recording selected in the left list. The display shows level meters with peak hold and clip indicators.

### AUTO-SAMPLING AN INSTRUMENT

1. **Connect it.** In the standalone app open *Options -> Audio/MIDI settings*: pick the MIDI output that plays your synth and the audio input it comes back on. (In a DAW, route the plug-in's MIDI out to the synth and the synth's audio into the plug-in - not every DAW allows MIDI out of an instrument, so the app is easiest.) Modular synths can use the CV/gate on Out 3/4.
2. **Plan it** on the AUTO-SAMPLER page: start and end note, note step, number of velocity layers (spread between AS Min and AS Max velocity), round robins, note length (AS Coarse seconds + AS Fine ms) and AS Gap (time left for the release).
3. **Press TEST.** A note plays and the status line shows the measured latency in samples. Set *AS Latency* a little below it so each sample starts right at the attack.
4. **Press AUTO.** Every note × velocity × round robin is played and recorded into a new *AUTOnnnn* folder, already named for SD Multisample (e.g. 060_C4_V1_RR1.wav). Press AUTO again to stop early.
5. **Play it.** Click the folder in the right-hand *Auto-sampled* list and the plug-in switches to SD Multisample with it loaded.

# PARAMETERS

| # | Parameter | Range | Default | Page | Values / notes |
|---|---|---|---|---|---|
| 1 | Attenuverter 1 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 2 | Attenuverter 2 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 3 | Attenuverter 3 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 4 | Attenuverter 4 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 5 | Attenuverter 5 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 6 | Attenuverter 6 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 7 | Input gain | 0 .. 24 dB | 0 dB | RECORD | |
| 8 | Sample rate | 0 .. 1 | 48kHz | RECORD | 0=48kHz, 1=96kHz |
| 9 | Bit depth | 0 .. 1 | 16 bit | RECORD | 0=16 bit, 1=24 bit |
| 10 | Channels | 1 .. 6 | 2 | RECORD | |
| 11 | Rec CV Mode | 0 .. 1 | Trigger | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 12 | Play CV Mode | 0 .. 1 | Trigger | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 13 | Play As Loop | 0 .. 1 | One-shot | RECORD | 0=One-shot, 1=Loop |
| 14 | Outputs 1/2 | 0 .. 2 | Thru/Play | RECORD | 0=Thru, 1=Thru/Play, 2=Play |
| 15 | Outputs 3/4 | 0 .. 2 | Thru | RECORD | 0=Thru, 1=Thru/Play, 2=Play |
| 16 | AS Enable | 0 .. 1 | Off | AUTO-SAMPLER | 0=Off, 1=On |
| 17 | AS Start Note | 0 .. 127 | 21 | AUTO-SAMPLER | |
| 18 | AS End Note | 0 .. 127 | 108 | AUTO-SAMPLER | |
| 19 | AS Note Step | 1 .. 127 | 1 | AUTO-SAMPLER | |
| 20 | AS Vel Sw | 1 .. 9 | 1 | AUTO-SAMPLER | |
| 21 | AS RRobins | 1 .. 9 | 1 | AUTO-SAMPLER | |
| 22 | AS Coarse | 0 .. 60 s | 1 s | AUTO-SAMPLER | |
| 23 | AS Fine | -1000 .. 1000 ms | 0 ms | AUTO-SAMPLER | |
| 24 | AS Gap | 1 .. 600 | 1 | AUTO-SAMPLER | |
| 25 | AS Latency | 0 .. 960 | 0 | AUTO-SAMPLER | |
| 26 | Thru routing 1 | 0 .. 6 | 1 | RECORD | 0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4 |
| 27 | Thru routing 2 | 0 .. 6 | 2 | RECORD | 0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4 |
| 28 | Thru routing 3 | 0 .. 6 | None | display | 0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4 |
| 29 | Thru routing 4 | 0 .. 6 | None | display | 0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4 |
| 30 | Thru routing 5 | 0 .. 6 | None | display | 0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4 |
| 31 | Thru routing 6 | 0 .. 6 | None | display | 0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4 |
| 32 | AS Min velocity | 0 .. 127 | 25 | AUTO-SAMPLER | |
| 33 | AS Max velocity | 0 .. 127 | 127 | AUTO-SAMPLER | |
| 34 | Playback file | 0 .. 9999 | 0 | list | |
| 35 | Record enable | 0 .. 1 | On | display | 0=Locked, 1=On |

| # | Parameter | Range | Default | Page | Values / notes |
|---|---|---|---|---|---|
| 36 | Playback enable | 0 .. 1 | On | display | 0=Locked, 1=On |
| 37 | Playback gain | -40 .. 6 dB | 0 dB | RECORD | |
| 38 | AS Preview note | 0 .. 127 | 48 | AUTO-SAMPLER | |

# EX POLY WAVETABLE




An 8-voice wavetable synthesizer. Pick a wavetable in the left list; the right list shows its waves - click one to jump the *Wave offset* to it. The position can also follow velocity, pitch, both envelopes and the LFO, and *Wave spread* gives every voice a different position. Each voice has two envelopes (with curve shapes; envelope 2's sustain can be negative), a state-variable filter (low/band/high pass) and an LFO; unison stacks up to 8 voices per note.

- **OSC** wave offset/spread, unison, voices, gain · **ENV 1**, **ENV 2** · **FILTER** type, cutoff (as a MIDI note), resonance and modulation · **MOD** routings and LFO speed/retrigger/spread · **CHORD** · **FX** Juno-style chorus I/II, delay, output mode (*Per voice* sends voices to Out 1-4) · **KEYS** tuning, sustain, trigger mode (gated, drone, triggered), Scala.
- Wavetables are band-limited (mip-mapped) so high notes stay clean.

## YOUR OWN WAVETABLES AND WAVEEDIT

- **Drag and drop** WAV files or folders onto the window, or press **IMPORT**.
- **WAVEEDIT** imports every bank saved by the WaveEdit app in *Documents/WaveEdit/banks*; press it again after editing to update them.
- WaveEdit banks (64 waves × 256 samples) and disting tables (600-sample cycles) are recognised automatically; a set of single-cycle files becomes one table. Each import becomes a folder of single-cycle WAVs in *wavetables/*, so it also works on a real disting EX.

# PARAMETERS

| # | Parameter | Range | Default | Page | Values / notes |
|---|---|---|---|---|---|
| 1 | Attenuverter 1 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 2 | Attenuverter 2 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 3 | Attenuverter 3 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 4 | Attenuverter 4 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 5 | Attenuverter 5 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 6 | Attenuverter 6 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 7 | Wavetable | 0 .. 999 | 0 | list | |
| 8 | Wave offset | -100 .. 100 | 0 | OSC | |
| 9 | Wave spread | -100 .. 100 | 0 | OSC | |
| 10 | Coarse tune | -60 .. 60 | 0 | KEYS | |
| 11 | Fine tune | -100 .. 100 cents | 0 cents | KEYS | |
| 12 | Attack time | 0 .. 127 | 20 | ENV 1 | |
| 13 | Decay time | 0 .. 127 | 60 | ENV 1 | |
| 14 | Sustain level | 0 .. 127 | 80 | ENV 1 | |
| 15 | Release time | 0 .. 127 | 60 | ENV 1 | |
| 16 | Attack shape | 0 .. 127 | 64 | ENV 1 | |
| 17 | Decay shape | 0 .. 127 | 64 | ENV 1 | |
| 18 | Attack time 2 | 0 .. 127 | 60 | ENV 2 | |
| 19 | Decay time 2 | 0 .. 127 | 70 | ENV 2 | |
| 20 | Sustain level 2 | -127 .. 127 | 64 | ENV 2 | |
| 21 | Release time 2 | 0 .. 127 | 50 | ENV 2 | |
| 22 | Attack shape 2 | 0 .. 127 | 64 | ENV 2 | |
| 23 | Decay shape 2 | 0 .. 127 | 64 | ENV 2 | |
| 24 | Filter type | 0 .. 3 | Off | FILTER | 0=Off, 1=Lowpass, 2=Bandpass, 3=Highpass |
| 25 | Filter freq | 0 .. 127 | 64 | FILTER | |
| 26 | Filter Q | 0 .. 100 | 50 | FILTER | |
| 27 | Veloc -> volume | 0 .. 100 % | 100 % | MOD | |
| 28 | Veloc -> wave | -100 .. 100 % | 0 % | MOD | |
| 29 | Veloc -> filter | -127 .. 127 | 0 | FILTER | |
| 30 | Pitch -> wave | -100 .. 100 % | 0 % | MOD | |
| 31 | Pitch -> filter | -100 .. 100 % | 0 % | FILTER | |
| 32 | Env -> wave | -100 .. 100 % | 0 % | MOD | |
| 33 | Env -> filter | -127 .. 127 | 0 | FILTER | |
| 34 | Env 2 -> wave | -100 .. 100 % | 0 % | MOD | |
| 35 | Env 2 -> filter | -127 .. 127 | 0 | FILTER | |
| 36 | Env 2 -> pitch | -120 .. 120 | 0 | MOD | |
| 37 | LFO -> wave | -100 .. 100 % | 0 % | MOD | |
| 38 | LFO -> filter | -127 .. 127 | 0 | FILTER | |
| 39 | LFO -> pitch | -120 .. 120 | 0 | MOD | |
| 40 | LFO speed | -100 .. 100 | 90 | MOD | |
| 41 | Gain | -40 .. 24 dB | 0 dB | OSC | |
| 42 | Sustain | 0 .. 1 | Off | KEYS | 0=Off, 1=On |
| 43 | Bend range | 0 .. 48 | 2 | KEYS | |

| # | Parameter | Range | Default | Page | Values / notes |
|---|---|---|---|---|---|
| 44 | Chord enable | 0 .. 7 | Off | CHORD | 0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3 |
| 45 | Chord key | -12 .. 12 | 0 | CHORD | |
| 46 | Chord scale | 0 .. 7 | Major | CHORD | 0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic |
| 47 | Chord shape | 0 .. 13 | None | CHORD | 0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth |
| 48 | Chord inversion | 0 .. 3 | 0 | CHORD | |
| 49 | Arpeggio 1 mode | 0 .. 14 | Off | display | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |
| 50 | Arpeggio 2 mode | 0 .. 14 | Off | display | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |
| 51 | Arpeggio 3 mode | 0 .. 14 | Off | CHORD | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |
| 52 | Arpeggio 1 range | 1 .. 3 | 1 | display | |
| 53 | Arpeggio 2 range | 1 .. 3 | 1 | display | |
| 54 | Arpeggio 3 range | 1 .. 3 | 1 | CHORD | |
| 55 | Scala/MTS | -16 .. 99 | 0 | KEYS | |
| 56 | Scala KBM | 0 .. 99 | 0 | KEYS | |
| 57 | Chorus mode | 0 .. 2 | Off | FX | 0=Off, 1=I, 2=II |
| 58 | Delay mode | 0 .. 2 | Off | FX | 0=Off, 1=Stereo, 2=Ping-pong |
| 59 | Delay level | -40 .. 0 dB | -3 dB | FX | |
| 60 | Delay time | 1 .. 10000 ms | 500 ms | FX | |
| 61 | Delay feedback | 0 .. 100 % | 50 % | FX | |
| 62 | Unison | 1 .. 8 | 1 | OSC | |
| 63 | Unison detune | 0 .. 100 cents | 10 cents | OSC | |
| 64 | Output spread | -100 .. 100 % | 0 % | FX | |
| 65 | Output mode | 0 .. 3 | Spread voice | FX | 0=Spread voice, 1=Spread voice 2, 2=Spread pitch, 3=Per voice |
| 66 | Input mode | 0 .. 18 | 1 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 67 | Sustain mode | 0 .. 1 | Synth | KEYS | 0=Synth, 1=Piano |
| 68 | MIDI vel curve | 0 .. 3 | Linear | KEYS | 0=Linear, 1=Soft, 2=Hard, 3=Fixed |
| 69 | Wave input | 0 .. 6 | 5 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 70 | Pitch bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 71 | Arp reset input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 72 | Max voices | 1 .. 8 | 8 | OSC | |
| 73 | LFO retrigger | 0 .. 2 | Poly | MOD | 0=Poly, 1=Mono, 2=Off |
| 74 | LFO spread | 0 .. 90 deg | 0 deg | MOD | |
| 75 | Gate offset | 0 .. 496 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 76 | Trigger mode | 0 .. 2 | Gated | KEYS | 0=Gated, 1=Drone, 2=Triggered |
| 77 | Break time | 0 .. 1000 ms | 0 ms | CHORD | |
| 78 | Break direction | 0 .. 2 | Up | CHORD | 0=Up, 1=Down, 2=Alternate |

# EX POLY FM




An 8-voice, 4-timbre FM synthesizer using the six-operator engine of Mutable Instruments Plaits - the same engine the disting EX uses. Banks (left list) are DX7 32-voice SysEx files from the card's *FMSYX* folder; the right list shows the bank's voices by name. The TIMBRE buttons choose which timbre the lists set; timbres 2-4 can follow timbre 1.

- **T1 … T4**: output, gain, pan, *Brightness* (modulator depth) and *Env scale* (envelope times) per timbre.
- **VOICE**: max voices, unison, master gain, velocity -> brightness, voice transpose and output mode. **CHORD**, **FX** (chorus, delay), **KEYS**.
- Parameters 54-64 are not listed in the module's manual; ex-girlfriend uses them for the voice and effect extras.

## EDITING WITH A DX7 EDITOR

Point a DX7 editor/librarian (Dexed, Patch Base, MIDI Quest …) at the plug-in. Single-voice dumps, parameter edits and dump requests apply to the timbre chosen by *SysEx target*; the display then shows **VOICE EDITED**. From MENU -> Poly FM: *Write voice* stores the edit in the bank (in memory), *Write bank to SD* saves it, *Abandon edit* reverts, and *Send* dumps the voice or bank back to the editor. *Write built-in to SD* creates FMSYX with the built-in banks.

# PARAMETERS

| # | Parameter | Range | Default | Page | Values / notes |
|---|---|---|---|---|---|
| 1 | Attenuverter 1 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 2 | Attenuverter 2 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 3 | Attenuverter 3 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 4 | Attenuverter 4 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 5 | Attenuverter 5 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 6 | Attenuverter 6 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 7 | Bank 1 | 1 .. 999 | 1 | list | |
| 8 | Voice 1 | 1 .. 32 | 1 | list | |
| 9 | Output 1 | 0 .. 7 | 1/2 | T1 | 0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4 |
| 10 | Gain 1 | -40 .. 24 dB | 0 dB | T1 | |
| 11 | Pan 1 | -100 .. 100 % | 0 % | T1 | |
| 12 | Brightness 1 | -100 .. 100 % | 0 % | T1 | |
| 13 | Env scale 1 | -100 .. 100 % | 0 % | T1 | |
| 14 | Bank 2 | 0 .. 999 | 0 | list | |
| 15 | Voice 2 | 0 .. 32 | 0 | list | |
| 16 | Output 2 | 0 .. 7 | 1/2 | T2 | 0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4 |
| 17 | Gain 2 | -40 .. 24 dB | 0 dB | T2 | |
| 18 | Pan 2 | -100 .. 100 % | 0 % | T2 | |
| 19 | Brightness 2 | -100 .. 100 % | 0 % | T2 | |
| 20 | Env scale 2 | -100 .. 100 % | 0 % | T2 | |
| 21 | Bank 3 | 0 .. 999 | 0 | list | |
| 22 | Voice 3 | 0 .. 32 | 0 | list | |
| 23 | Output 3 | 0 .. 7 | 1/2 | T3 | 0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4 |
| 24 | Gain 3 | -40 .. 24 dB | 0 dB | T3 | |
| 25 | Pan 3 | -100 .. 100 % | 0 % | T3 | |
| 26 | Brightness 3 | -100 .. 100 % | 0 % | T3 | |
| 27 | Env scale 3 | -100 .. 100 % | 0 % | T3 | |
| 28 | Bank 4 | 0 .. 999 | 0 | list | |
| 29 | Voice 4 | 0 .. 32 | 0 | list | |
| 30 | Output 4 | 0 .. 7 | 1/2 | T4 | 0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4 |
| 31 | Gain 4 | -40 .. 24 dB | 0 dB | T4 | |
| 32 | Pan 4 | -100 .. 100 % | 0 % | T4 | |
| 33 | Brightness 4 | -100 .. 100 % | 0 % | T4 | |
| 34 | Env scale 4 | -100 .. 100 % | 0 % | T4 | |
| 35 | Coarse tune | -60 .. 60 ST | 0 ST | KEYS | |
| 36 | Fine tune | -100 .. 100 cents | 0 cents | KEYS | |
| 37 | Sustain | 0 .. 1 | Off | KEYS | 0=Off, 1=On |
| 38 | Bend range | 0 .. 48 | 2 | KEYS | |
| 39 | Chord enable | 0 .. 7 | Off | CHORD | 0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3 |
| 40 | Chord key | -12 .. 12 | 0 | CHORD | |

| # | Parameter | Range | Default | Page | Values / notes |
|---|---|---|---|---|---|
| 41 | Chord scale | 0 .. 7 | Major | CHORD | 0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic |
| 42 | Chord shape | 0 .. 13 | None | CHORD | 0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth |
| 43 | Chord inversion | 0 .. 3 | 0 | CHORD | |
| 44 | Arpeggio 1 mode | 0 .. 14 | Off | display | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |
| 45 | Arpeggio 2 mode | 0 .. 14 | Off | display | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |
| 46 | Arpeggio 3 mode | 0 .. 14 | Off | CHORD | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |
| 47 | Arpeggio 1 range | 1 .. 3 | 1 | display | |
| 48 | Arpeggio 2 range | 1 .. 3 | 1 | display | |
| 49 | Arpeggio 3 range | 1 .. 3 | 1 | CHORD | |
| 50 | Scala/MTS | -16 .. 99 | 0 | KEYS | |
| 51 | Scala KBM | 0 .. 99 | 0 | KEYS | |
| 52 | Output mode | 0 .. 2 | Spread voice | VOICE | 0=Spread voice, 1=Spread voice 2, 2=Spread pitch |
| 53 | Output spread | -100 .. 100 % | 0 % | VOICE | |
| 54 | Max voices | 1 .. 8 | 8 | VOICE | |
| 55 | Unison | 1 .. 8 | 1 | VOICE | |
| 56 | Unison detune | 0 .. 100 cents | 10 cents | VOICE | |
| 57 | Chorus mode | 0 .. 2 | Off | FX | 0=Off, 1=I, 2=II |
| 58 | Delay mode | 0 .. 2 | Off | FX | 0=Off, 1=Stereo, 2=Ping-pong |
| 59 | Delay level | -40 .. 0 dB | -3 dB | FX | |
| 60 | Delay time | 1 .. 10000 ms | 500 ms | FX | |
| 61 | Delay feedback | 0 .. 100 % | 50 % | FX | |
| 62 | Gain | -40 .. 24 dB | 0 dB | VOICE | |
| 63 | Veloc -> bright | -100 .. 100 % | 0 % | VOICE | |
| 64 | Voice transpose | 0 .. 1 | Use | VOICE | 0=Ignore, 1=Use |
| 65 | Input mode | 0 .. 9 | 1 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 66 | Sustain mode | 0 .. 1 | Synth | KEYS | 0=Synth, 1=Piano |
| 67 | MIDI vel curve | 0 .. 3 | Linear | KEYS | 0=Linear, 1=Soft, 2=Hard, 3=Fixed |
| 68 | Pitch bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 69 | Arp reset input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 70 | Gate offset | 0 .. 496 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 71 | Trigger mode | 0 .. 2 | Gated | KEYS | 0=Gated, 1=Drone, 2=Triggered |
| 72 | Break time | 0 .. 1000 ms | 0 ms | CHORD | |
| 73 | Break direction | 0 .. 2 | Up | CHORD | 0=Up, 1=Down, 2=Alternate |
| 74 | SysEx target | 1 .. 4 | 1 | KEYS | |

# EX POLY EXCITER




An 8-voice synthesizer that excites a resonating string with a short burst of sound - plucked, struck, scraped or bowed. The left list holds exciter tables (*Internal*, or tables from the card's *exciters* folder); the right list shows the exciters in the table. Drum hits make great exciters: a kick for a soft tone, a snare for a sharp, noisy attack.

| Mode | Behaviour |
|---|---|
| Strike | The exciter plays once, for the whole Exciter length. |
| Scrape | As Strike, but the exciter stops when the key is released. |
| Strum | The exciter loops while the key is held; its level follows **pressure** (aftertouch or the Pressure parameter) - silent without it. |

| Internal exciter | Sound |
|---|---|
| 0 Ramp | A decaying linear ramp. |
| 1 Noise burst | White noise at a constant level. |
| 2 Noise ramp | Noise shaped by a decaying ramp. |
| 3 Square | Constant level: strong plucks at the start and end. |
| 4 Silence | For processing external audio only. |

- **EXCITER** mode, exciter, length, pressure, *Ext exciter input* (feed audio into the plug-in to excite the strings live).
- **RESONATOR** tone (brightness), damping, extra damping on release, low-pass filter. **VELO** velocity/pitch/pressure modulation. **ENV/LFO** modulate pitch and exciter choice. **CHORD**, **FX**, **KEYS**.

# PARAMETERS

| # | Parameter | Range | Default | Page | Values / notes |
|---|---|---|---|---|---|
| 1 | Attenuverter 1 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 2 | Attenuverter 2 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 3 | Attenuverter 3 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 4 | Attenuverter 4 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 5 | Attenuverter 5 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 6 | Attenuverter 6 | -200 .. 200 % | 100 % | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 7 | Mode | 0 .. 2 | Strike | EXCITER | 0=Strike, 1=Scrape, 2=Strum |
| 8 | Table | -1 .. 999 | -1 | list | |
| 9 | Exciter | 0 .. 127 | 0 | EXCITER | |
| 10 | Exciter length | 1 .. 597 ms | 100 ms | EXCITER | |
| 11 | Tone | 0 .. 127 | 50 | RESONATOR | |
| 12 | Damp | 0 .. 100 | 4 | RESONATOR | |
| 13 | Release damp | 0 .. 100 | 0 | RESONATOR | |
| 14 | Pressure | 0 .. 127 | 0 | EXCITER | |
| 15 | Veloc -> strike | 0 .. 100 % | 100 % | VELO | |
| 16 | Veloc -> tone | -100 .. 100 % | 0 % | VELO | |
| 17 | Veloc -> damp | -100 .. 100 % | 0 % | VELO | |
| 18 | Veloc -> exciter | -127 .. 127 | 0 | VELO | |
| 19 | Pitch -> damp | -100 .. 100 % | 0 % | VELO | |
| 20 | Pitch -> exciter | -100 .. 100 % | 0 % | VELO | |
| 21 | Pitch -> LPF | -100 .. 100 % | 0 % | RESONATOR | |
| 22 | Press -> exciter | -127 .. 127 | 0 | VELO | |
| 23 | Env -> pitch | -120 .. 120 | 0 | ENV/LFO | |
| 24 | Env -> exciter | -127 .. 127 | 0 | ENV/LFO | |
| 25 | LFO -> pitch | -200 .. 200 cents | 0 cents | ENV/LFO | |
| 26 | LFO -> exciter | -127 .. 127 | 0 | ENV/LFO | |
| 27 | LFO speed | -100 .. 100 | 90 | ENV/LFO | |
| 28 | Attack time | 0 .. 127 | 60 | ENV/LFO | |
| 29 | Decay time | 0 .. 127 | 70 | ENV/LFO | |
| 30 | Sustain level | -127 .. 127 | 0 | ENV/LFO | |
| 31 | Release time | 0 .. 127 | 50 | ENV/LFO | |
| 32 | Coarse tune | -60 .. 60 | 0 | KEYS | |
| 33 | Fine tune | -100 .. 100 cents | 0 cents | KEYS | |
| 34 | Gain | -40 .. 24 dB | 0 dB | RESONATOR | |
| 35 | Sustain | 0 .. 1 | Off | KEYS | 0=Off, 1=On |
| 36 | Bend range | 0 .. 48 | 2 | KEYS | |
| 37 | Chord enable | 0 .. 7 | Off | CHORD | 0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3 |
| 38 | Chord key | -12 .. 12 | 0 | CHORD | |
| 39 | Chord scale | 0 .. 7 | Major | CHORD | 0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic |

| # | Parameter | Range | Default | Page | Values / notes |
|---|---|---|---|---|---|
| 40 | Chord shape | 0 .. 13 | None | CHORD | 0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth |
| 41 | Chord inversion | 0 .. 3 | 0 | CHORD | |
| 42 | Arpeggio 1 mode | 0 .. 14 | Off | display | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |
| 43 | Arpeggio 2 mode | 0 .. 14 | Off | display | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |
| 44 | Arpeggio 3 mode | 0 .. 14 | Off | CHORD | 0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played |
| 45 | Arpeggio 1 range | 1 .. 3 | 1 | display | |
| 46 | Arpeggio 2 range | 1 .. 3 | 1 | display | |
| 47 | Arpeggio 3 range | 1 .. 3 | 1 | CHORD | |
| 48 | Scala/MTS | -16 .. 99 | 0 | KEYS | |
| 49 | Scala KBM | 0 .. 99 | 0 | KEYS | |
| 50 | Low pass filter | 0 .. 127 | 126 | RESONATOR | |
| 51 | Chorus mode | 0 .. 2 | Off | FX | 0=Off, 1=I, 2=II |
| 52 | Delay mode | 0 .. 2 | Off | FX | 0=Off, 1=Stereo, 2=Ping-pong |
| 53 | Delay level | -40 .. 0 dB | -3 dB | FX | |
| 54 | Delay time | 1 .. 10000 ms | 500 ms | FX | |
| 55 | Delay feedback | 0 .. 100 % | 50 % | FX | |
| 56 | Unison | 1 .. 8 | 1 | FX | |
| 57 | Unison detune | 0 .. 100 cents | 10 cents | FX | |
| 58 | Output spread | -100 .. 100 % | 0 % | FX | |
| 59 | Break time | 0 .. 1000 ms | 0 ms | CHORD | |
| 60 | Break direction | 0 .. 2 | Up | CHORD | 0=Up, 1=Down, 2=Alternate |
| 61 | Output mode | 0 .. 3 | Spread voice | FX | 0=Spread voice, 1=Spread voice 2, 2=Spread pitch, 3=Per voice |
| 62 | Input mode | 0 .. 18 | 1 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 63 | Sustain mode | 0 .. 1 | Synth | KEYS | 0=Synth, 1=Piano |
| 64 | MIDI vel curve | 0 .. 3 | Linear | KEYS | 0=Linear, 1=Soft, 2=Hard, 3=Fixed |
| 65 | Ext exciter input | 0 .. 6 | None | EXCITER | 0=None, 1=In L, 2=In R, 3=In L+R, 4=In L+R, 5=In L+R, 6=In L+R |
| 66 | Pitch bend input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 67 | Arp reset input | 0 .. 6 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |
| 68 | Max voices | 1 .. 8 | 8 | KEYS | |
| 69 | LFO retrigger | 0 .. 1 | Poly | ENV/LFO | 0=Poly, 1=Off |
| 70 | LFO spread | 0 .. 90 deg | 0 deg | ENV/LFO | |
| 71 | Gate offset | 0 .. 496 | 0 | - | *CV/hardware only - kept for numbering, no effect in the plug-in* |

# TROUBLESHOOTING

| Problem | Fix |
|---|---|
| macOS says the app is damaged or from an unidentified developer; the DAW skips the plug-ins | The builds are not notarised. Run the `xattr -dr com.apple.quarantine …` command from the install chapter, then rescan plug-ins. |
| No sound from EX Multisample | Check the SD folder has sample folders (SD -> Show in Finder) and that a folder is selected in the left list. |
| Notes hang | Press PANIC (or MIDI -> All notes off). |
| The DAW does not see Out 3/4 | Enable the plug-in's second output bus in the DAW's routing/IO settings. |
| Auto-sampled files start late or early | Run TEST and set AS Latency just below the measured value. |
| Large folders take a moment to load | Samples load into memory in the background; very large libraries (hundreds of MB) take a few seconds. |
| A wavetable change does nothing | On the display, Wavetable needs confirming (PUSH V): press Esc, or choose the table in the list instead. |

# CREDITS AND LICENCE

- ex-girlfriend is open source under the **GNU AGPL-3.0**; source code at github.com/femifleming/ex-girlfriend.
- Built with JUCE 9 (AGPLv3). Poly FM uses the FM engine of Mutable Instruments Plaits by Émilie Gillet (MIT).
- Behaviour follows the disting EX user manual by Expert Sleepers. Not affiliated with Expert Sleepers.
- Sample library: VSCO-2 Community Edition, Karoryfer (Emily Guitar, Growlybass, Hadzi-Fia), Virtuosity Drums, AKWF wavetables, Greg Sullivan CP80 (CC BY 3.0) and generated content - see CREDITS.txt in the library for details.